

	Environmental Analysis Teaching and Research Laboratory	Date: 6/16/2025	SOP No. X
	Standard Operating Procedure	Title: Nitrate and Ammonium	
	Approved By: Los Huertos	Revised: June 30, 2025	

1. Scope and Application

2. Health and Safety

3. Personnel & Training Responsibilities

Students using this SOP should be trained for the following SOPsa:

- SOP1
- SOP2

4. Required Materials

4.1 Reciprocating shaker

4.2 Graduated cylinder or repipet type dispenser

4.3 Filter funnels

4.4 Whatman 42 filter paper

4.5 Paper bags or aluminum weigh boats

4.6 Specimen containers, solo cups or nalgene bottles

4.7 Sample cups and caps

4.8 Pipet

4.9 Top-loading balance

5. Estimated Time

5.1 This will take XX minutes...

6. Lab Procedure

6.1 Preparation: Add 298.2 g CaCl_2 to 2 liters distilled water

6.2 Fill volumetric container w/ 30 ml solution

Table 1: Soil Texture Correction Factor

Texture	Moist Soil	Dry Soil
Sand	2.3	2.6
Loam	2.0	2.5
Clay	1.7	2.2

6.3 Add soil into container until solution reaches 40 ml

6.4 cap and shake until all soil is well mixed

6.5 allow sample to sit and soil particles to settle (around 10 mins, no more than 1hr)

6.6 Dip test strip in clearer solution near top for one second

6.7 compare strip to standard color chart at 30 and 60 seconds

6.8 Interpretation of N testing quick strips

6.9 Calculate correction factor using soil textures (average values) See Texture Analysis SOP.

6.10 Corrected value = test strip value of N (ppm) / correction factor

6.11 record nitrate ppm and correction factor in field notebook

7. References

7.1 APHA, AWWA. WEF. (2012) Standard Methods for examination of water and wastewater. 22nd American Public Health Association (Eds.). Washington. 1360 pp. (2014).